

Predicting Clinical Trial Duration for Pharmaceutical Investment Analysis

A Regression Tree Approach Using the AACT Clinical Trials Database

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Introduction

Pharmaceutical royalty investors provide upfront capital in exchange for a share of a drug's future sales revenue. The value of a royalty depends on the timing of future cash flows; a drug that is approved and reaches the market sooner delivers a more valuable royalty stream than a drug that is delayed. This project aims to predict clinical trial duration for pharmaceutical royalty investment analysis by building a regression tree model, using characteristics known at the time a trial is registered, to provide an estimate that could complement potential royalty investment evaluations. We will also compare the effects of hyperparameter tuning and tree restrictions on Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Data

The data was sourced from the Aggregate Analysis of ClinicalTrials.gov (AACT) database, a relational PostgreSQL database maintained by the Clinical Trials Transformation Initiative. A local PostgreSQL instance was set up and populated with a static AACT snapshot. Features were extracted via SQL queries joining across the studies, sponsors, facilities, conditions, interventions, countries, designs, and eligibilities tables. Based off the provided data dictionary (https://aact.ctti-clinicaltrials.org/data_dictionary), the following features were used:

- nct_id: Unique trial identifier
- phase: Trial phase (1, 2, 3, 4)
- enrollment: Target number of participants a trial aims to recruit
- overall_status: Trial completion status
- start_date: Start of trial

- completion_date: End of trial
- agency_class: Trial lead sponsor type
- num_sites: How many locations the trial ran across
- num_conditions: Number of distinct conditions/diseases the trial targets
- num_interventions: Number of distinct treatment arms
- num_countries: Number of countries the trial ran across
- allocation: Randomized vs. non-randomized design
- intervention_model: Type of intervention model
- masking: Number of parties blinded
- primary_purpose: Purpose of trial
- gender, minimum_age, maximum_age, healthy_volunteers: Eligibility criteria

We used phase/enrollment/sponsor as core structural drivers, num_sites/num_countries for geographic complexity, and design/eligibility fields as hypothesized secondary drivers. Feature selection excluded any column only knowable after a trial begins or is completed, avoiding data leakage. We computed the target variable, duration_days, as the total length of the trials using the variables start_date and completion_date, which are then dropped from the dataset.

Preprocessing

The following scoping decisions were made:

- Restricted the data to overall_status = 'COMPLETED' trials, since both start and end dates are required to compute the target variable.
- Restricted the data to study_type = 'INTERVENTIONAL' trials, since observational studies do not have a defined phase and are not the relevant population for a project focused on drug development timelines.
- Excluded trials with phase recorded as null, scoping the analysis to trials following the standard phase-based development pathway.

Missing values were handled differently depending on its apparent cause. Rows with small fractions of missing values in enrollment, allocation, intervention_model, masking, primary_purpose, gender, and healthy_volunteers were dropped entirely from the dataset since the cost of doing so is small. The fields minimum_age and maximum_age were passed directly to the chosen model, LightGBM, which handles missing values during split finding. These

missing values reflect a trial with no upper age limit specified in its eligibility criteria, which is valuable information and is treated accordingly.

LightGBM will be able to isolate outlier-prone splits during tree construction, avoiding the need for filtering for outliers. The final modelling dataset contained approximately 114,000 completed interventional trials.

Modelling

Unlike settings where the number of predictors approaches the number of observations, this dataset has a comparatively large sample size relative to its feature count ($n \gg p$), so dimensionality reduction is not strictly necessary here. Ridge regression and LASSO apply an L2 or L1 penalty to a linear model's coefficients to control variance and, in LASSO's case, perform variable selection by shrinking some coefficients to zero. Both remain fundamentally linear/additive models, however, and neither directly captures non-linear relationships or interaction effects among predictors. Regression trees address this by recursively partitioning the feature space, allowing the model to capture non-linearities and interactions without requiring them to be pre-specified. This motivated the use of a tree-based ensemble method (LightGBM) as the primary modelling approach, with linear regression, incorporating the same features via median imputation and one-hot encoding, as a baseline for direct comparison.

At each node, a regression tree selects the split (feature and threshold) that minimizes the resulting within-node variance, or equivalently maximizes variance reduction between the parent and child nodes. LightGBM builds an ensemble of such trees sequentially, with each new tree fit to the residual errors of the current ensemble (gradient boosting), and predictions given by the sum of all trees' outputs. The hyperparameters tuned in this project (num_leaves and max_depth, which bound the size and depth of each tree, and min_child_samples, which sets a minimum number of observations per leaf) play a role analogous to the shrinkage parameter λ in ridge or LASSO regression: larger, less-constrained trees fit training data more closely but risk overfitting, while more restrictive settings trade fit for generalization. This is the bias-variance tradeoff.

We set up a chronological train-test split instead of a traditional random split since clinical trial regulatory requirements and timelines have shifted over the multi-decade span of the dataset, splitting on the date January 1st 2019 to ensure an 80-20 split of the data. LightGBM models were fit before and after a log transform was applied to the target variable to address the right-skewedness of `duration_days`. Compared to the base model, the log-transformed model reduced MAE from 428.91 to 335.05 and RMSE from 570.83 to 463.09 (Table 1).

A model with hyperparameter tuning was performed via `RandomizedSearchCV` with a `TimeSeriesSplit` cross-validation (5 folds), respecting the chronological ordering established rather than using a standard k-fold cross-validation. Another model with early stopping was also evaluated as an alternative preventative against potential overfitting. A final model combining both hyperparameter tuning and early stopping was fitted.

Table 1. Model performance across modelling decisions.

Model	MAE	RMSE
Raw target	428.91	570.83
Log-transformed target	335.05	463.09
+ hyperparameter tuning	337.52	469.65
+ early stopping	336.30	467.60
+ hyperparameter tuning and early stopping	325.33	444.02

Results

The LightGBM model with both hyperparameter tuning and early stopping performed the best out of the models. Compared with the linear regression model, it reduced MAE from 377.27 to 325.33 and RMSE from 640.28 to 444.02 (Table 2). SHAP summary plot shows that sponsor type had a significant impact on prediction, along with `healthy_volunteers`.

Table 2. Best model comparison with linear regression.

Model	MAE	RMSE
Linear regression	377.27	640.28
LightGBM (log-transformed target, hyperparameter tuning, early stopping)	325.33	444.02

The large gap in RMSE indicates that the linear regression model makes significantly larger errors than the tree-based model avoids, consistent with the non-linear relationships between trial characteristics and duration that a linear, additive model cannot represent.

SHAP (Shapley Additive Explanations) values were computed for the final model to interpret which features drive its predictions and in which directions (Figure 1). The most influential features, in order of importance, were `sponsor_type`, `healthy_volunteers`, `enrollment`, `num_sites`, `num_countries`, and `maximum_age`. Larger target enrollment consistently pushed predicted duration upward, consistent with recruitment time scaling with the number of participants needed. Trials run across more sites and more countries were predicted to take longer, plausibly reflecting the coordination and regulatory overhead of operating across more locations and jurisdictions. Trials with a higher or unbounded maximum eligible age tended to have longer predicted durations, a less intuitive finding. Trials open to healthy volunteers, most commonly early-phase safety studies, showed a distinct effect on predicted duration relative to trials restricted to participants with the condition under study, plausibly reflecting simpler recruitment and screening for a healthy volunteer population.

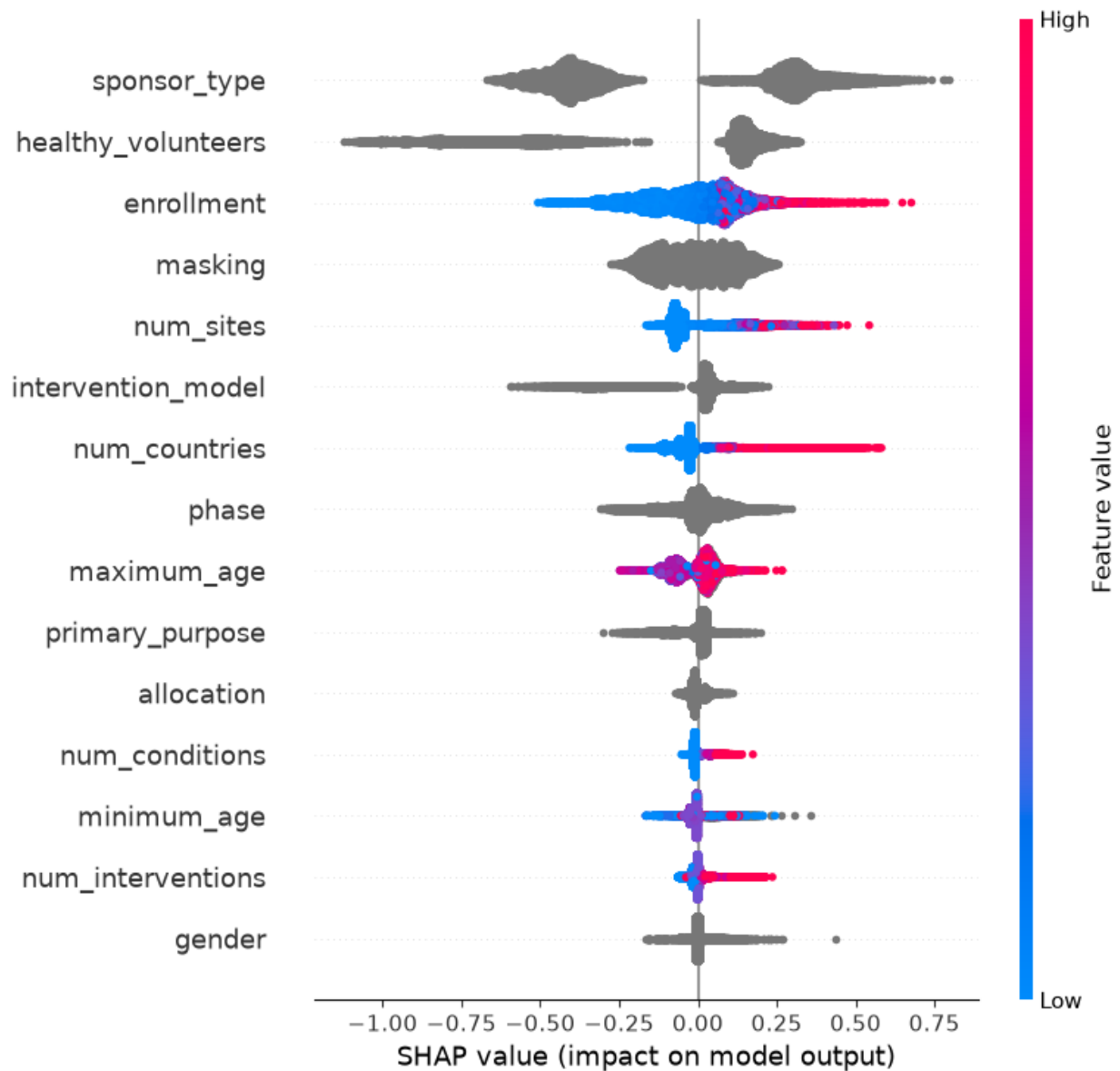


Figure 1. SHAP values for LightGBM with hyperparameter tuning and early stopping.

Applications

To connect the model to its intended investment application, its output was used in a toy discounted cash flow illustration. In a pharmaceutical royalty deal, an investor pays an upfront amount for the right to a future royalty stream, contingent on the drug's approval and sales. Since that royalty payment only arrives after the trial completes, its value today is discounted for the time the investor must wait.

For an illustrative large, multi-site, multi-country Phase 3 trial (3,000 participants, 150 sites, 20 countries), the model predicted a duration of approximately 2.3 years. Applying a 10% discount rate to a future \$100 million royalty payment over this predicted duration yields a present value of \$80.2 million (i.e., the maximum an investor should be willing to pay today for that future payment, given this timing estimate). The model's output may help inform a feature-driven estimate of how much a given future payment is worth in the moment, which can be used to support a negotiated upfront cost.

Limitations

- The model is trained exclusively on trials that reached completion. This means the model does not address trial failure risks and its duration estimates are implicitly conditional on the trial completing at all.
- The data used only contains registry-level features, so drug-specific or company-specific factors are not captured.

Conclusion

This project developed a LightGBM regression tree model with hyperparameter tuning and early stopping, predicting clinical trial duration from characteristics available at trial registration, achieving a meaningful improvement over a linear regression model baseline (MAE 325.33 vs. 377.27; RMSE 444.02 vs. 640.28). SHAP analysis provided interpretable reasoning for individual predictions. The model's output was connected to a pharmaceutical royalty investment use case, informing the timing assumptions that determine the present value of a contingent future royalty payment.

Appendix

The code for this project can be found in the GitHub repository: <https://github.com/eth-tran/Pharmaceutical-Investment-Analysis>

Data obtained from: <https://aact.ctti-clinicaltrials.org/downloads>